

Auteur: Rune Suy
Studierichting: Informatica
Oefeningenreeks 5L nr. 1.1

$$f(x) = \frac{4}{3}x \text{ over } [0, 3]$$

$$f'(x) = \frac{4}{3} \Rightarrow f'(x)^2 = \frac{16}{9} \Rightarrow f'(x)^2 + 1 = \frac{25}{9}$$

$$S = 2\pi \int_0^3 \left| \frac{4}{3}x \right| \sqrt{\frac{25}{9}} dx = \frac{20}{9}\pi [x^2]_0^3 = 20\pi$$

5L-1.1-rune.suy.INF.png

References